King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 208 Final Exam Term 241

Original exam date: 19 December 2024. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

---

1. If $y(x)$ is the solution of the initial-value problem

$$\frac{dy}{dx} = 4x^3 y - y, \qquad y(1) = -3,$$

then $y(0) =$

(a) 0

(b) -2

(c) 4

(d) 5

(e) -3

2. The guaranteed radius of convergence of the power series solution of the differential equation $(x^2+4)y'' + 4xy' + 2y = 0$ about the ordinary point $x = 0$ is

(a) 2

(b) $\sqrt{2}$

(c) 0

(d) 4

(e) ∞

3. The rank of the matrix

$$\begin{bmatrix} 1 & 1 & -1 & 7 \\ 1 & 4 & 5 & 16 \\ 1 & 3 & 3 & 13 \\ 2 & 5 & 4 & 23 \end{bmatrix}$$

is

(a) 4

(b) 2

(c) 3

(d) 1

(e) 0

4. If $y = \sum_{n=0}^{\infty} c_n x^n$ is a power series solution about the ordinary point $x = 0$ of the differential equation $y'' + x^2 y = 0$, then the coefficients c_n satisfy

(a) $c_{n+4} = \dfrac{-c_n}{(n+3)(n+4)},\ n \geq 0$

(b) $c_{n+4} = \dfrac{3c_n}{(n+1)(n+4)},\ n \geq 0$

(c) $c_{n+4} = \dfrac{2c_{n+1}}{(n+1)(n+3)},\ n \geq 0$

(d) $c_{n+4} = \dfrac{-c_{n+1}}{(n+1)(n+3)},\ n \geq 0$

(e) $c_{n+4} = \dfrac{2c_n}{(n+1)(n+4)},\ n \geq 0$

5. The general solution of the system $X' = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} X$ is given by

(a) $X = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{4t}$

(b) $X = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 3 \\ 0 \end{bmatrix} e^{4t}$

(c) $X = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{4t}$

(d) $X = c_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{4t}$

(e) $X = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{4t}$

6. For $A = \begin{bmatrix} 7 & -6 \\ 12 & -10 \end{bmatrix}$, if P is a diagonalizing matrix such that $P^{-1}AP = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$, then

(a) $P = \begin{bmatrix} 4 & 7 \\ 3 & 2 \end{bmatrix}$

(b) $P = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$

(c) $P = \begin{bmatrix} 4 & 6 \\ 3 & 5 \end{bmatrix}$

(d) $P = \begin{bmatrix} 1 & 7 \\ 3 & 5 \end{bmatrix}$

(e) $P = \begin{bmatrix} 4 & 7 \\ 2 & 5 \end{bmatrix}$

7. A possible fundamental matrix of the system $X' = \begin{bmatrix} 9 & 5 \\ -6 & -2 \end{bmatrix} X$ is

(a) $\Phi(t) = \begin{bmatrix} 5e^{3t} & e^{4t} \\ e^{3t} & 2e^{4t} \end{bmatrix}$

(b) $\Phi(t) = \begin{bmatrix} 5e^{3t} & e^{4t} \\ -6e^{3t} & -e^{4t} \end{bmatrix}$

(c) $\Phi(t) = \begin{bmatrix} 5e^{3t} & e^{4t} \\ -6e^{3t} & 2e^{4t} \end{bmatrix}$

(d) $\Phi(t) = \begin{bmatrix} e^{3t} & e^{4t} \\ -6e^{3t} & -e^{4t} \end{bmatrix}$

(e) $\Phi(t) = \begin{bmatrix} 4e^{3t} & e^{4t} \\ e^{3t} & -e^{4t} \end{bmatrix}$

8. Let $F(t) = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$, $e^{At} = \begin{bmatrix} 1+3t & -t \\ 9t & 1-3t \end{bmatrix}$. A particular solution for the system $X' = AX + F(t)$ is

(a) $X_p = \begin{bmatrix} 6t^2 - 5t \\ 24t^2 + 4t \end{bmatrix}$

(b) $X_p = \begin{bmatrix} 13t^2 + 2t \\ -5t \end{bmatrix}$

(c) $X_p = \begin{bmatrix} 6t^2 + 2t \\ 24t^2 - 5t \end{bmatrix}$

(d) $X_p = \begin{bmatrix} 13t^2 - 2t \\ t^2 - 5t \end{bmatrix}$

(e) $X_p = \begin{bmatrix} 8t^2 + 7t \\ 24t^2 + 5t \end{bmatrix}$

9. The solution of the Bernoulli equation

$$y^2 y' + 2xy^3 = 6x$$

is

(a) $y^2 = 3 + ce^{-3x^2}$

(b) $y^3 = 5 + ce^{-3x^2}$

(c) $y^4 = 6 + ce^{-3x^2}$

(d) $y^2 = 5 + ce^{-3x^2}$

(e) $y^3 = 3 + ce^{-3x^2}$

10. If the recurrence relation of the Frobenius series solution of the differential equation $2xy'' + (1+x)y' + y = 0$ that corresponds to the indicial root $r = \dfrac{1}{2}$ is given by

$$c_{k+1} = \frac{-c_k}{2(k+1)}, \qquad k = 0, 1, 2, \ldots,$$

then the first three terms in the solution are given by

(a) $x^{\frac{1}{2}} - \dfrac{1}{2}x^{\frac{3}{2}} + x^{\frac{5}{2}}$

(b) $x^{\frac{1}{2}} + \dfrac{1}{3}x^{\frac{3}{2}} + \dfrac{1}{8}x^{\frac{5}{2}}$

(c) $x^{\frac{1}{2}} - \dfrac{1}{3}x^{\frac{3}{2}} + 4x^{\frac{5}{2}}$

(d) $x^{\frac{1}{2}} + \dfrac{1}{4}x^{\frac{3}{2}} - 5x^{\frac{5}{2}}$

(e) $x^{\frac{1}{2}} - \dfrac{1}{2}x^{\frac{3}{2}} + \dfrac{1}{8}x^{\frac{5}{2}}$

11. The largest indicial root at $x = 0$ for the differential equation $4xy'' + 2y' + y = 0$ is

(a) $r = \dfrac{1}{3}$

(b) $r = 2$

(c) $r = 3$

(d) $r = 0$

(e) $r = \dfrac{1}{2}$

12. The Wronskian of the functions

$$y_1 = e^{2x}, \quad y_2 = e^{4x}$$

is

(a) $3e^{6x}$

(b) $7e^{6x}$

(c) $5e^{6x}$

(d) $2e^{6x}$

(e) e^{6x}

13. The solution of the exact differential equation

$$(3x^2 + 2y^2)\,dx + (4xy + 6y^2)\,dy = 0$$

is given by

(a) $x^3 + 2xy^2 + 2y^3 = c$

(b) $x^2 + 2xy^3 + 2y^2 = c$

(c) $x^3 - xy^2 + 2y^3 = c$

(d) $x^3 + 4xy^3 + 2y^2 = c$

(e) $x^3 - 2xy^2 + 4y^3 = c$

14. If A is a 2×2 real matrix with an eigenvector $\begin{bmatrix} 1+2i \\ 5 \end{bmatrix}$ associated with the eigenvalue $\lambda = 5+2i$ of A, then the general solution of the system $X' = AX$ is

(a) $X = c_1 \begin{bmatrix} \cos(5t) - 2\sin(5t) \\ 5\cos(5t) \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 2\cos(5t) + \sin(5t) \\ 5\sin(5t) \end{bmatrix} e^{5t}$

(b) $X = c_1 \begin{bmatrix} \cos(5t) \\ \sin(5t) \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} \sin(5t) \\ \cos(5t) \end{bmatrix} e^{2t}$

(c) $X = c_1 \begin{bmatrix} \cos(2t) - 2\sin(2t) \\ 5\cos(2t) \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} \sin(3t) \\ 5\sin(3t) \end{bmatrix} e^{5t}$

(d) $X = c_1 \begin{bmatrix} \cos(2t) - 2\sin(2t) \\ 5\cos(2t) \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 2\cos(2t) + \sin(2t) \\ 5\sin(2t) \end{bmatrix} e^{5t}$

(e) $X = c_1 \begin{bmatrix} \cos(2t) \\ 5\cos(2t) \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 2\cos(2t) + 2\sin(2t) \\ 5\sin(2t) \end{bmatrix} e^{5t}$

15. An appropriate form of particular solution of the differential equation $y^{(3)} + y' = 2 - \sin x$ is

(a) $y_p = A + Bx\cos x + Cx\sin x$

(b) $y_p = Ax + B\cos x + C\sin x$

(c) $y_p = Ax + Bx\cos x + Cx\sin x$

(d) $y_p = Ax^2 + B\cos x + C\sin x$

(e) $y_p = Ax^2 + Bx\cos x + Cx\sin x$

16. The general solution of the differential equation

$$D(D-2)^2(D^2-4D+13)\,y=0$$

is given by

(a) $y = c_1 + c_2e^{2x} + c_3xe^{2x} + c_4e^{3x}\cos(4x) + c_5e^{3x}\sin(4x)$

(b) $y = c_1 + c_2e^{2x} + c_3xe^{2x} + c_4e^{2x}\cos(2x) + c_5e^{2x}\sin(2x)$

(c) $y = c_1 + c_2e^{2x} + c_3xe^{2x} + c_4e^{2x}\cos(3x) + c_5e^{2x}\sin(3x)$

(d) $y = c_1 + c_2e^{2x} + c_4e^{2x}\cos(3x) + c_5e^{2x}\sin(3x)$

(e) $y = c_1 + c_2e^{2x} + c_3xe^{2x} + c_4\cos(3x) + c_5\sin(3x)$

17. The matrix $A = \begin{bmatrix} 5 & -1 & -1 \\ 3 & -2 & -6 \\ -2 & 5 & 9 \end{bmatrix}$ has a defective eigenvalue $\lambda = 4$ of defect 2. If we choose $V_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ such that $(A-4I)^3V_3 = \mathbf{0}$ and $(A-4I)^2V_3 \neq \mathbf{0}$, then the general solution of the system $X' = AX$ is

(a) $X = \left(c_1 \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} t \\ 3t \\ 3t-2 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 3t \\ \frac{3}{2}t^2 - 2t \end{bmatrix} \right) e^{4t}$

(b) $X = \left(c_1 \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 1+t \\ -3t+3 \\ 3t \end{bmatrix} + c_3 \begin{bmatrix} 1+t \\ t^2 \\ t \end{bmatrix} \right) e^{4t}$

(c) $X = \left(c_1 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -3t+3 \\ 3t-2 \end{bmatrix} + c_3 \begin{bmatrix} t+1 \\ -\frac{3}{2}t^2 + 3t \\ \frac{3}{2}t^2 - 2t \end{bmatrix} \right) e^{4t}$

(d) $X = \left(c_1 \begin{bmatrix} 1 \\ -3 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 3 \\ 3t-2 \end{bmatrix} + c_3 \begin{bmatrix} t \\ 2t^2 + 3t \\ t^2 - 2t \end{bmatrix} \right) e^{4t}$

(e) $X = \left(c_1 \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -3t+3 \\ 3t-2 \end{bmatrix} + c_3 \begin{bmatrix} t+1 \\ -\frac{3}{2}t^2 + 3t \\ \frac{3}{2}t^2 - 2t \end{bmatrix} \right) e^{4t}$

18. For the nilpotent matrix $A = \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix}$, $e^{At} =$

(a) $\begin{bmatrix} 2t & -2t \\ t & 1-2t \end{bmatrix}$

(b) $\begin{bmatrix} 1+3t & -2t \\ t & 1-2t \end{bmatrix}$

(c) $\begin{bmatrix} 1-2t & t \\ 2t & 1+4t \end{bmatrix}$

(d) $\begin{bmatrix} 1+2t & -2t \\ 2t & 1-2t \end{bmatrix}$

(e) $\begin{bmatrix} 1+2t & t \\ 4t & -2t \end{bmatrix}$

19. The largest eigenvalue of the matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ -2 & 3 & -1 \\ -6 & 6 & 0 \end{bmatrix}$ is $\lambda =$

(a) 4

(b) 5

(c) 2

(d) 3

(e) 1

20. Let $\mathbf{w} = (1, 5, 8)$; $\mathbf{v}_1 = (-1, 2, 3)$, $\mathbf{v}_2 = (3, 1, -2)$, $\mathbf{v}_3 = (2, 3, 0)$ be four vectors in $\mathbb{R}^3$. If $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, then $a + b + c =$

(a) 10

(b) 8

(c) 5

(d) 9

(e) 7

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(e)**	$y = Ce^{x^4 - x}$, $y(1) = C = -3$, so $y(0) = C = -3$.
2	**(a)**	$x^2 + 4 = 0$ at $x = \pm 2i$, distance 2 from 0: radius 2.
3	**(b)**	Row reduction leaves two nonzero rows: rank 2.
4	**(a)**	Coefficient of x^{n+2}: $(n+4)(n+3)c_{n+4} + c_n = 0$, so $c_{n+4} = \frac{-c_n}{(n+3)(n+4)}$.
5	**(a)**	$\lambda = -1,\ 4$ with eigenvectors $(1, -1)$ and $(3, 2)$.
6	**(b)**	Eigenvectors $(3, 4)$ for -1 and $(2, 3)$ for -2: $P = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$.
7	**(b)**	$\lambda = 3, 4$ with eigenvectors $(5, -6)$, $(1, -1)$: columns $\mathbf{v}_i e^{\lambda_i t}$.
8	**(e)**	$X_p = e^{At} \int e^{-As} F\, ds = e^{At} \begin{bmatrix} 7t - 8t^2 \\ 5t - 24t^2 \end{bmatrix} = \begin{bmatrix} 8t^2 + 7t \\ 24t^2 + 5t \end{bmatrix}$.
9	**(e)**	$v = y^3$: $v' + 6xv = 18x$, $\rho = e^{3x^2}$, so $y^3 = 3 + ce^{-3x^2}$.
10	**(e)**	$c_0 = 1$, $c_1 = -\frac{1}{2}$, $c_2 = \frac{1}{8}$: $x^{1/2} - \frac{1}{2}x^{3/2} + \frac{1}{8}x^{5/2}$.
11	**(e)**	Indicial equation $4r(r-1) + 2r = 0 \Rightarrow r = 0,\ \frac{1}{2}$; largest is $\frac{1}{2}$.
12	**(d)**	$W = e^{2x} \cdot 4e^{4x} - 2e^{2x} \cdot e^{4x} = 2e^{6x}$.
13	**(a)**	$F = x^3 + 2xy^2 + g(y)$, $g' = 6y^2$: $x^3 + 2xy^2 + 2y^3 = c$.
14	**(d)**	$\mathbf{a} = (1, 5)$, $\mathbf{b} = (2, 0)$: $\mathbf{x}_1 = e^{5t}(\mathbf{a}\cos 2t - \mathbf{b}\sin 2t)$, $\mathbf{x}_2 = e^{5t}(\mathbf{b}\cos 2t + \mathbf{a}\sin 2t)$.
15	**(c)**	$y_c = c_1 + c_2\cos x + c_3 \sin x$, so both groups need a factor x: $y_p = Ax + Bx\cos x + Cx\sin x$.
16	**(c)**	Roots $0,\ 2,\ 2,\ 2 \pm 3i$: $y = c_1 + c_2 e^{2x} + c_3 x e^{2x} + e^{2x}(c_4\cos 3x + c_5 \sin 3x)$.
17	**(e)**	Chain $V_3 = (1, 0, 0)$, $V_2 = (1, 3, -2)$, $V_1 = (0, -3, 3)$; solutions $V_1 e^{4t}$, $(V_2 + tV_1)e^{4t}$, $(V_3 + tV_2 + \frac{t^2}{2}V_1)e^{4t}$.
18	**(d)**	$A^2 = 0$, so $e^{At} = I + At = \begin{bmatrix} 1 + 2t & -2t \\ 2t & 1 - 2t \end{bmatrix}$.
19	**(d)**	$\det(A - \lambda I) = -\lambda(\lambda - 1)(\lambda - 3)$, so the largest eigenvalue is 3.
20	**(e)**	Solving the system gives $a = 6,\ b = 5,\ c = -4$, so $a + b + c = 7$.